Final Review

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ECE 3340 Exam 4

ECE 3340 Final Exam

EXAM RULES:

- DO NOT open or start the exam until told to do so by the instructor.
- This exam is open-book and open-notes. You are allowed to use a regular (graphical) calculator, but **NOT** allowed to use a computer/tablet, a smart phone, or a calculator app on a smart phone.
- If you have any questions, ask the instructor. You will not be given credit for work that is based on a wrong assumption.
- Please sit separately, don't be too close to each other.
- If you finish the exam before the time is up, raise your hand and the instructor will collect your exam. You may then quietly leave after your exam is submitted.
- When time is called, you must stop all writing and put your pens/pencils down. Stay seated until all exams have been collected. The instructor will notify you when you may leave.
- You **MUST** sign the academic honesty agreement below before taking the exam. Failure to do so will result in a **zero** on the exam.

ACADEMIC HONESTY AGREEMENT:

I Understand and agree to abide by the provisions in the **University of Houston Undergraduate Academic Honesty Policy**. I promise that the work on this exam is my own and that I have received no help during this exam other than any assistance provided by the instructor and/or TAs in this course. I understand that academic honesty is taken very seriously, and in the cases of violations, penalties may include suspension or expulsion from the University of Houston.

Name (print): ______________________ MYUH# ____________

Signature: ______________________

Note:

- **Please show your work for all critical steps, do NOT just provide an answer**. If you use calculator to solve the problem, please indicate it.
- If you use a method different from the one specified in the problem, provide detailed explanations of how you solve it.
- For trigonometric operations, inputs are **radians**, not degrees.

1. (Binary Floating Point) Assume an 8-bit floating point format with a 4-bit exponent and a bias of 7 and implied 1.
   - Provide the decimal representations of the following stored values.

Sign	Exponent				Mantissa		
1	0	0	1	1	0	1	0

Sign	Exponent				Mantissa		
0	1	0	1	0	1	1	0

   - For this format, if the spacings between any two adjacent floating points are the same? If your answer is yes, please provide this spacing in decimal value. If your answer is no, please provide the largest and smallest spacings in decimal values.

2. (Horner's Algorithm) Calculate $f(x) = \frac{\cos(x-1)}{\sqrt{x}}$ for $x = 1.2$ using Taylor Series Expansion and Horner's algorithm to an error of $O(x^4)$ and 3 digits of precision. Demonstrate Horner's algorithm using synthetic division. Show your work, including all derivations.

3. (LU Decomposition) Use partial pivoting gaussian elimination to factorize the following matrix $\boldsymbol{A}$. Show your work by filling in the following matrices to find the lower triangular matrix $\boldsymbol{L}$, the upper triangular matrix $\boldsymbol{U}$, and the permutation matrix $\boldsymbol{P}$, which satisfy $\boldsymbol{PA} = \boldsymbol{LU}$.

$$\boldsymbol{A} = \begin{bmatrix} \frac{3}{2} & -\frac{25}{3} & \frac{21}{2} \\ -3 & 2 & -5 \\ 1 & -\frac{35}{3} & \frac{32}{3} \end{bmatrix}$$

$$\boldsymbol{P} = \begin{bmatrix} \quad \end{bmatrix}, \boldsymbol{L} = \begin{bmatrix} \quad \end{bmatrix}, \boldsymbol{U} = \begin{bmatrix} \quad \end{bmatrix}$$

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4. (Roots of equations) Use the combined Newton's and Bisection methods to find the root for the following function for 3 iterations given the interval $[-3, 3]$ and $x_0 = 0$. For each iteration, you need to update the interval $[a, b]$ and/or the root x_i. Try to use Newton's method as possible. Draw the line representing each c_n or x_n. Label the line with (1) the iteration n, (2) the x-axis position, (3) the interval side (a or b) that c_n replaces (if needed), and (4) the procedure to determine x_{n+1}. Please show your work using the following templates of bisection and newton's methods and graphically illustrate the procedures on the following figure.

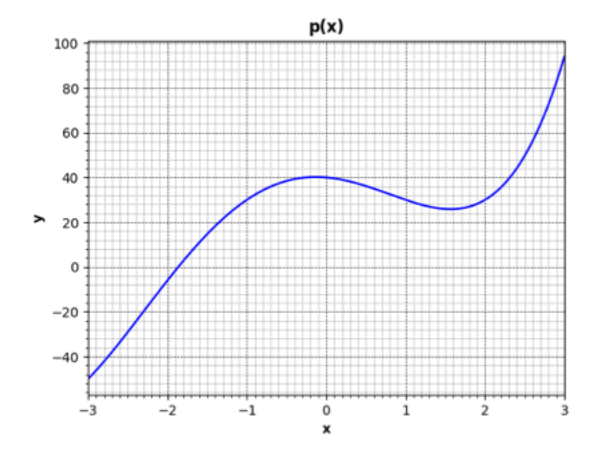
$$p(x) = x^4 + 3x^3 - 11x^2 - 3x + 40 = 0$$

Bisection:

a $\quad p(a)$ $\quad b$ $\quad p(b)$ $\quad c$ $\quad p(c)$

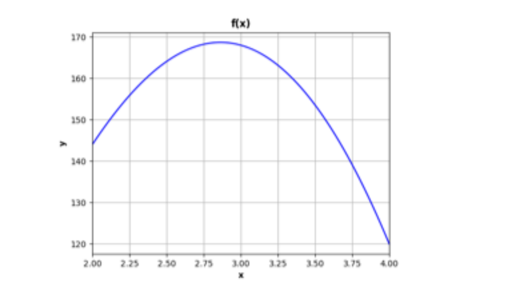
Newton:

x_n $\quad p(x_n)$ $\quad p'(x_n)$ $\quad x_{n+1}$



5. **(Numerical Integration)** Approximate the following integral using Simpson's rule and the Trapezoid rule with **4** intervals. Use the same **five** points for the Simpson's rule and Trapezoid rule. Show your work. (Hint: For Simpson's rule, you can use $\int_a^b f(x)dx = \int_a^{\frac{a+b}{2}} f(x)dx + \int_{\frac{a+b}{2}}^b f(x)dx$.)

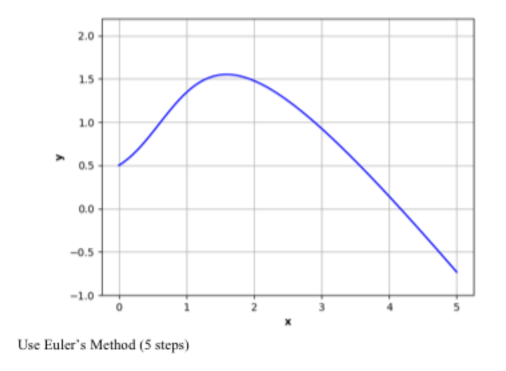
$$\int_2^4 f(x)dx$$



6. **(Explicit Integration)** Solve the following differential equation using explicit integration:

$$f(x, y) = \frac{dy}{dx} = \sin(x + y)$$

given the initial condition $(x, y) = (0, 0.5)$ and $\Delta x = 1.0$. **Plot** the resulting function alongside the real result in the figure below. You need to plot all the calculated (x_i, y_i) in the figure and draw the slope $(\frac{dy}{dx})$ at each (x_i, y_i).



Use Euler's Method (5 steps)

x $\quad y$ $\quad f(x, y)$

7. **(Interpolation)** An engineer is trying to use Lagrange Polynomials to fit a set of measured data (x_i, y_i) in the following. Please
   a. Show the expressions of all the Lagrange Basis Polynomials of this interpolation in production form.
   b. Show the Lagrange interpolation of the data and predict the value of y at $x = -3.0$.
   c. Do you think that the value of y at $x = -3.0$ looks reasonable or not? Explain it in one sentence.

x	−2.0	−1.0	0	1
y	−0.1	−0.9	0.1	1.2

8. **(Least square)** Fit the following points to a cubic polynomial $p(x) = a + bx + cx^2 + dx^3$

x	−2.0	−1.0	0	1	2	3
y	1	22	31	27	19	19

please

- Find the set of linear equations that can minimize the cost function using least square fitting.
- Solve the linear equations using a calculator to get a, b, c and d.
- Predict the y at point $x = -3.0$ and point $x = 0.5$ respectively, using your cubic polynomial.